Day 5: Troubleshooting Network Services

Task 1: Simulate a DNS resolution failure on your local machine by editing the hosts file to point www.spotify.com to a wrong IP address, then try to ping the domain and document the error message you receive.

Procedure:

1. Open Notepad with Administrator privileges.
2. Click File → Open and navigate to:
   C:\Windows\System32\drivers\etc\
3. Change the file type from Text Documents to All Files and open the hosts file.
4. Add the following entry at the end of the file:

192.0.2.10 www.spotify.com

5. Save the file. This entry temporarily maps www.spotify.com to an incorrect IP address.
6. Open Command Prompt and execute:

ping www.spotify.com

7. Observe the IP address displayed and record the ping response or error message.

Explanation:

The hosts file is a local file used by Windows to map domain names to IP addresses. When an entry exists in this file, Windows can use it instead of performing a normal DNS lookup. Therefore, an incorrect entry can cause a domain to resolve to the wrong IP address.

Expected Result:

The ping command should attempt to communicate with 192.0.2.10 instead of the actual Spotify server IP. The request may time out or fail because this is a reserved documentation IP address.

Solution:

Remove the incorrect entry from the hosts file, save the changes, and execute:

ipconfig /flushdns

This clears the local DNS cache. Run the ping command again to verify that the incorrect mapping is no longer being used.

Task 2: Configure a DHCP server in a network simulator (such as Cisco Packet Tracer or GNS3) and intentionally set an incorrect default gateway; connect a client device and verify the assigned gateway, then describe how you would identify and fix this issue.

Objective:

To understand how an incorrectly configured default gateway in a DHCP server affects communication between devices and external networks.

Procedure:

1. Open Cisco Packet Tracer and create a simple network containing a router, switch, server, and client PC.
2. Configure the router's LAN interface with an IP address, for example, 192.168.1.1.
3. Configure the server with a DHCP pool containing an IP address range, subnet mask, DNS server, and default gateway.
4. Intentionally enter an incorrect default gateway, such as 192.168.1.254, in the DHCP configuration.
5. Open the client PC and select DHCP in its IP Configuration settings.
6. Open Command Prompt on the client and execute:

ipconfig /all

7. Observe the IP address, subnet mask, and default gateway assigned by the DHCP server.
8. Test communication with the router using:

ping 192.168.1.1

Explanation:

A default gateway is the router address used by a device to communicate with other networks. If DHCP provides an incorrect gateway, the client may receive a valid IP address but will not be able to forward traffic correctly to external networks.

Expected Result:

The client receives an IP address from the DHCP server but has an incorrect default gateway. Communication within the local subnet may work, while communication with external networks can fail.

Solution:

1. Open the DHCP server configuration.
2. Correct the default gateway to 192.168.1.1.
3. Save the DHCP configuration.
4. On the client PC, execute:

ipconfig /release

ipconfig /renew

5. Verify the updated gateway using ipconfig /all.
6. Test connectivity again using the ping command.

Task 3: Given a scenario where devices on your simulated network are unable to access websites by name but can by IP address, use the nslookup or dig tool to diagnose the DNS problem and write down the troubleshooting steps and solution.

Procedure:

1. Open Command Prompt on the client device.
2. Test basic network connectivity by pinging a known reachable IP address.
3. Check domain name resolution by executing:

nslookup www.google.com

4. Observe the DNS server address and the IP address returned by the command.
5. If the query fails or returns an unexpected result, check the configured DNS server using:

ipconfig /all

6. Verify that the configured DNS server is correct and reachable.
7. If the DNS server address is incorrect, update it to a valid DNS server address.
8. Clear the local DNS cache using:

ipconfig /flushdns

9. Run nslookup www.google.com again to verify successful name resolution.

Explanation:

DNS translates domain names into IP addresses. If DNS is not working correctly, users may be unable to access websites using names even when general IP connectivity is available.

Expected Result:

The nslookup command should return an IP address for the requested domain. Successful DNS resolution allows users to access websites using domain names.

Solution:

Check and correct the DNS server configuration, ensure that the DNS server is reachable, and verify that DNS traffic on port 53 is not blocked by a firewall.

Task 4: A device in your simulated network is not receiving an IP address from the DHCP server and shows an APIPA address (169.x.x.x). List the possible causes for this and document the step-by-step process you would use to resolve it.

Objective:

To understand why a client device receives an APIPA address instead of an IP address from the DHCP server and identify the steps required to resolve the problem.

What is APIPA?

APIPA stands for Automatic Private IP Addressing. Windows automatically assigns an IPv4 address from the range 169.254.0.0/16 when it cannot obtain an IP address from a DHCP server.

Possible Causes:

1. DHCP server is stopped or unavailable.
2. Network cable is disconnected or Wi-Fi is not connected.
3. DHCP scope has no available IP addresses.
4. Incorrect VLAN or network configuration.
5. Firewall is blocking DHCP communication.
6. DHCP relay configuration is missing or incorrect when the DHCP server is on another subnet.

Troubleshooting Procedure:

1. Check whether the device is properly connected to the network.
2. Execute ipconfig /all to verify whether the device has received an APIPA address.
3. Check whether the DHCP server is running and available.
4. Verify that the DHCP scope has sufficient available IP addresses.
5. Check the network cable, switch port, Wi-Fi connection, and VLAN configuration.
6. If the DHCP server is on another subnet, verify the DHCP relay configuration on the router.
7. Execute the following commands on the client:

ipconfig /release

ipconfig /renew

8. Finally, execute ipconfig /all to check whether a valid IP address, default gateway, and DNS server have been assigned.

Expected Result:

The client should receive a valid IP address from the DHCP server instead of an APIPA address. The device should then be able to communicate with other devices on the network.

Task 5: Create a table documenting at least three common DNS issues and three common DHCP issues you encountered or researched, including symptoms, possible causes, and recommended solutions for each.

Objective:

To identify common DNS and DHCP problems by documenting their symptoms, possible causes, and recommended troubleshooting solutions.

A. Common DNS Issues

DNS Issue	Symptoms	Possible Causes	Recommended Solutions
DNS Resolution Failure	Websites cannot be opened using domain names	Incorrect DNS server address or DNS service failure	Verify DNS settings and check DNS server availability
Incorrect DNS Record	Domain name resolves to the wrong IP address	Incorrect or outdated DNS record	Correct the DNS record and clear the DNS cache
DNS Server Unreachable	DNS queries time out or fail	Network connectivity problems or firewall blocking DNS traffic	Check network connectivity and allow DNS traffic on port 53

B. Common DHCP Issues

DHCP Issue	Symptoms	Possible Causes	Recommended Solutions
APIPA Address	Device receives an IP address starting with 169.254	DHCP server unavailable or DHCP request failure	Check server availability and renew the IP address
Incorrect Default Gateway	Device cannot communicate with external networks	Incorrect gateway configured in the DHCP scope	Correct the gateway and renew the DHCP lease
DHCP Scope Exhaustion	New devices cannot obtain an IP address	All available IP addresses have been assigned	Expand the DHCP scope or release unused leases

Important Troubleshooting Commands

Command	Purpose
ping	Tests network connectivity between devices
ipconfig /all	Displays complete IP configuration
ipconfig /release	Releases the current DHCP-assigned IP configuration
ipconfig /renew	Requests a new IP configuration from the DHCP server
ipconfig /flushdns	Clears the local DNS resolver cache
nslookup	Checks domain name resolution and DNS server responses